

SQA & Testing

Chapter 4

Software Quality

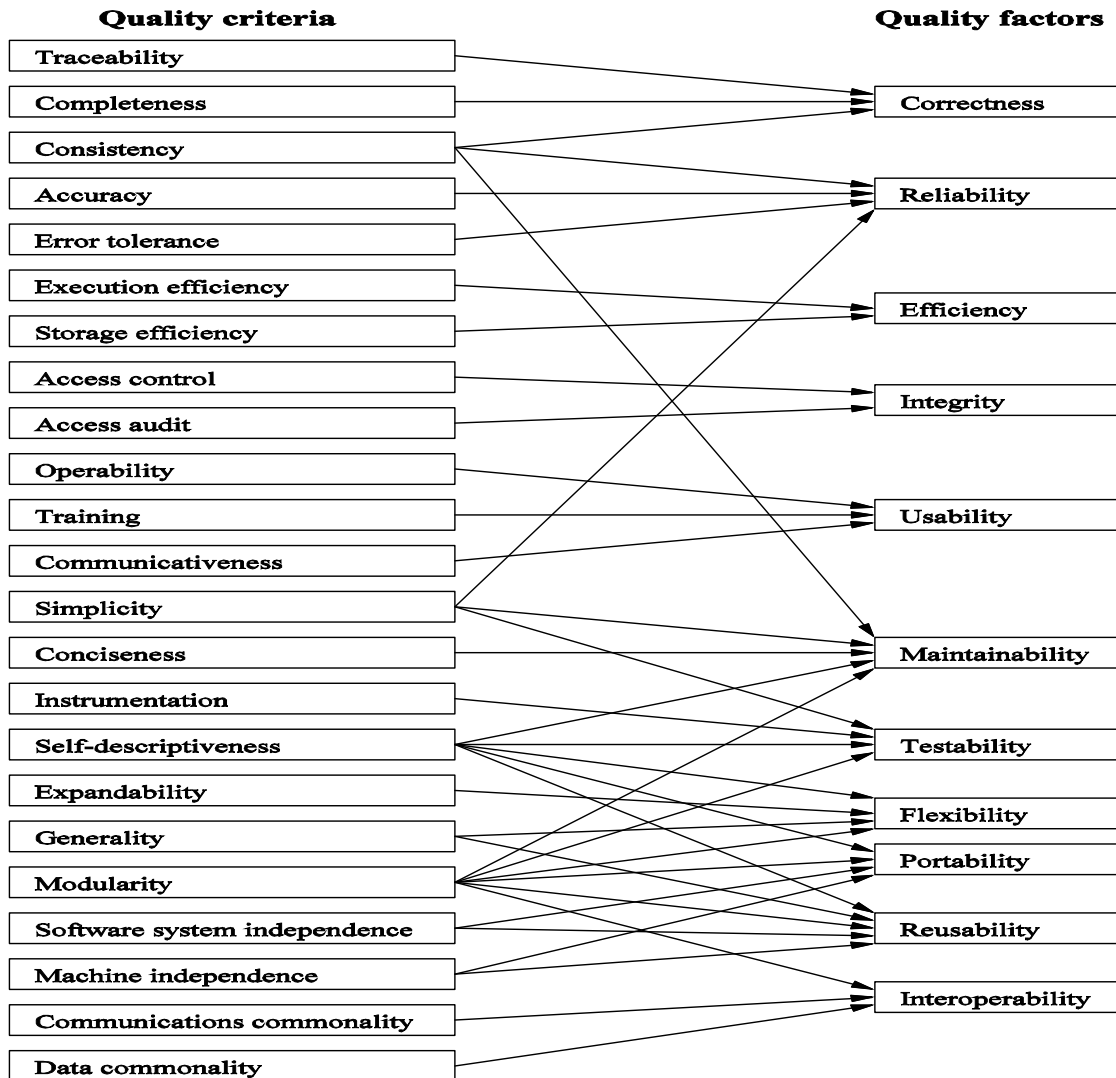
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Quality Factors	Definitions
Correctness	The extent to which a program satisfies its specifications and fulfills the user's mission objectives.
Reliability	The extent to which a program can be expected to perform its intended function with required precision.
Efficiency	The amount of computing resources and code required by a program to perform a function.
Integrity	The extent to which access to software or data by unauthorized persons can be controlled.
Usability	The effort required to learn, operate, prepare input, and interpret output of a program.
Maintainability	The effort required to locate and fix a defect in an operational program.
Testability	The effort required to test a program to ensure that it performs its intended functions.
Flexibility	The effort required to modify an operational program.
Portability	The effort required to transfer a program from one hardware and/ or software environment to another.
Reusability	The extent to which parts of a software system can be reused in other applications.
Interoperability	The effort required to couple one system with another.

Quality Categories	Quality Factors	Broad Objectives
Product Operation	Correctness Reliability Efficiency Integrity Usability	Does it do what the customer wants? Does it do it accurately all of the time? Does it quickly solve the intended problem? Is it secure? Can I run it?
Product Revision	Maintainability Testability Flexibility	Can it be fixed? Can it be tested? Can it be changed?
Product Transition	Portability Reusability Interoperability	Can it be used on another machine? Can parts of it be reused? Can it interface with another system?

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Quality Criteria	Definitions of Quality Criteria
Access audit	The ease with which software and data can be checked for compliance with standards or other requirements.
Access control	The provisions for control and protection of the software and data.
Accuracy	The precisions of computations and output.
Communication commonality	The degree to which standard protocols and interfaces are used.
Completeness	The degree to which a full implementation of the required functionalities has been achieved.
Communicativeness	The ease with which inputs and outputs can be assimilated.
Conciseness	The compactness of the source code, in terms of lines of code.
Consistency	The use of uniform design and implementation techniques and notations throughout a project.
Data commonality	The use of standard data representations.
Error tolerance	The degree to which continuity of operation is ensured under adverse conditions.
Execution efficiency	The run-time efficiency of the software.
Expandability	The degree to which storage requirements or software functions can be expanded.
Generality	The breadth of the potential application of software components.
Hardware independence	The degree to which the software is dependent on the underlying hardware.
Instrumentation	The degree to which the software provides for measurements of its use or identification of errors.
Modularity	The provision of highly independent modules.
Operability	The ease of operation of the software.
Self-documentation	The provision of in-line documentation that explains the implementation of components.
Simplicity	The ease with which the software can be understood.
Software system independence	The degree to which the software is independent of its software environment – non-standard language constructs, operating system, libraries, database management system, etc.
Software efficiency	The run-time storage requirements of the software.
Traceability	The ability to link software components to requirements.
Training	The ease with which new users can use the system.



Functionality: A set of attributes that bear on the existence of a set of functions and their specified properties. The functions are those that satisfy stated or implied needs.

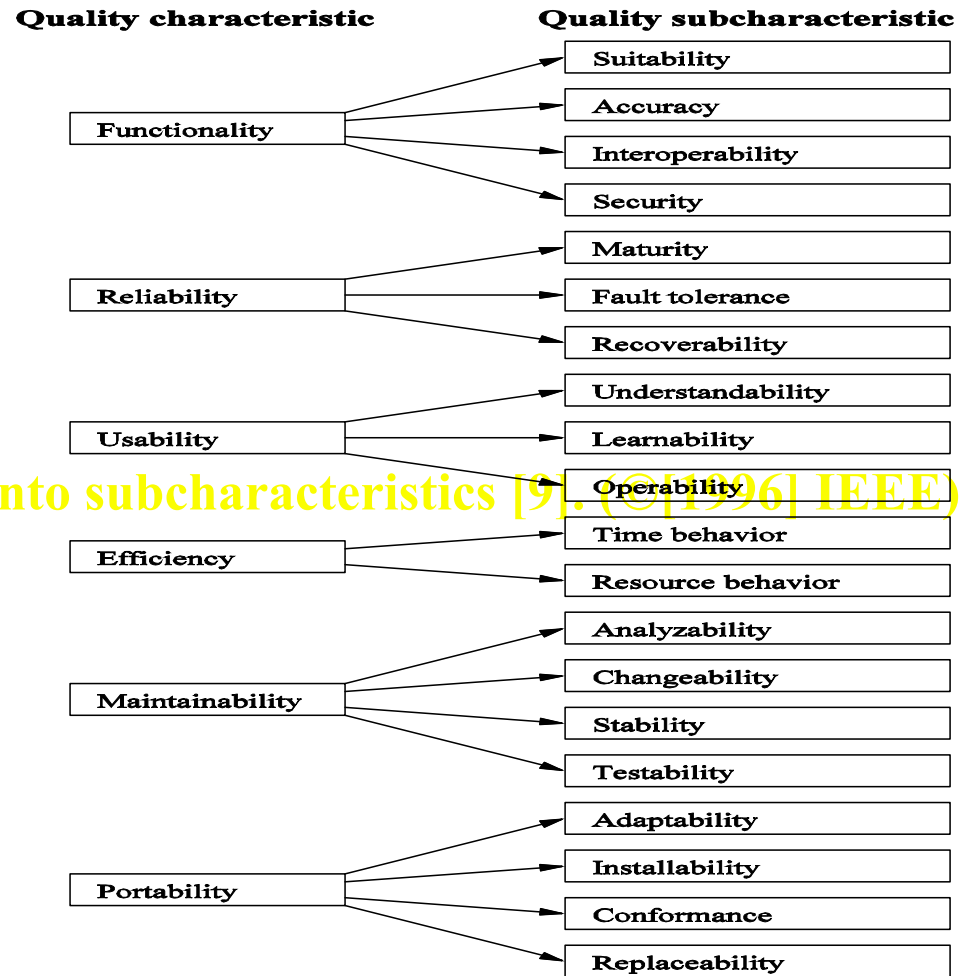
Reliability: A set of attributes that bear on the capability of software to maintain its performance level under stated conditions for a stated period of time.

Usability: A set of attributes that bear on the effort needed for use and on the individual assessment of such use by a stated or implied set of users.

Efficiency: A set of attributes that bear on the relationship between the software's performance and the amount of resource used under stated conditions.

Maintainability: A set of attributes that bear on the effort needed to make specified modifications (which may include corrections, improvements, or adaptations of software to environmental changes and changes in the requirements and functional specifications.)

Portability: A set of attributes that bear on the ability of software to be transferred from one environment to another (this includes the organizational, hardware or software environment.)



standard's features into subcharacteristics [9]. (©[1996] IEEE)

Suitability: The capability of the software to provide an adequate set of functions for specified tasks and user objectives.

Accuracy: The capability of the software to provide the right or agreed results or effects.

Interoperability: The capability of the software to interact with one or more specified systems.

Security: The capability of the software to prevent unintended access and resist deliberate attacks intended to gain unauthorized access to confidential information, or to make unauthorized modifications to information or to the program so as to provide the attacker with some advantage or so as to deny service to legitimate users.

Maturity: The capability of the software to avoid failure as a result of faults in the software.

Fault tolerance: The capability of the software to maintain a specified level of performance in case of software faults or of infringement of its specified interface.

Recoverability: The capability of the software to re-establish its level of performance and recover the data directly affected in the case of a failure.

Understandability: The capability of the software product to enable the user to understand whether the software is suitable, and how it can be used for particular tasks and conditions of use.

Learnability: The capability of the software product to enable the user to learn its applications.

Operability: The capability of the software product to enable the user to operate and control it.

Attractiveness: The capability of the software product to be liked by the user.

Time behavior: The capability of the software to provide appropriate response and processing times and throughput rates when performing its function under stated conditions.

Resource utilization: The capability of the software to use appropriate resources in an appropriate time when the software performs its function under stated condition.

Analyzability: The capability of the software product to be diagnosed for deficiencies or causes of failures in the software, or for the parts to be modified to be identified.

Changeability: The capability of the software product to enable a specified modification to be implemented.

Stability: The capability of the software to minimize unexpected effects from modifications of the software.

Testability: The capability of the software product to enable modified software to be validated.

Adaptability: The capability of the software to be modified for different specified environments without applying actions or means other than those provided for this purpose for the software considered.

Installability: The capability of the software to be installed in a specified environment.

Co-existence: The capability of the software to co-exist with other independent software in a common environment sharing common resources.

Replaceability: The capability of the software to be used in place of the other specified software in the environment of that software.

